

BANARAS HINDU UNIVERSITY

B.Voc. Semester I Examination, 2022-23

Paper: BVCA-104 — Skill Training in R

Time: Three Hours

Maximum Marks: 70

Instructions: Answer all Sections. Candidates are required to write their answers in their own words as far as practicable.

SECTION — A: Long Answer Questions

[15 × 2 = 30]

Instructions: Long answer type questions to be answered in about 750 words each.

Question 1. List out the different control structures in R Programming with general syntaxes and brief explanation. Write R code for the function using if-else command:

$$f(x, y) = \begin{cases} x + y & \text{if } x < y \\ 1 - x - y & \text{if } x > y \\ 0 & \text{if } x = y \end{cases}$$

OR

Question 2. Define nested if-else and nested loop statements with general syntaxes and explanation. Write R codes for finding a maximum number from the given three numbers.

Question 3. Write a brief note on R. Write R codes for computing matrix inverse for a 3×3 matrix.

OR

Question 4. Illustrate arithmetic, logical, and relational operators with an example for each. Write R program for the factorial function using a for-loop.

SECTION — B: Short Answer Questions

[5 × 6 = 30]

Instructions: Short answer type questions to be answered in about 550 words each. Attempt any six.

Question 1. Write general syntax of R functions that are used to create bar and pie charts. Show the percentage of students registered in Data Science, R Programming, and Machine Learning subjects using these charts. The percentages are 30%, 40%, and 30% in Data Science, R Programming, and Machine Learning, respectively.

Question 2. Giving a suitable example for each, explain the uses of the following operators:

`%x%`, `%*%`, `%%`, `%/%`, and `%in%`

Question 3. Using `function()` command, write R codes to enter the following mathematical expressions in R:

(a) $y = 10^x + \sqrt{x} \log_{10}(x)$

(b) $y = e^{x/2} - \sin(x) \tan(|x|^4)$

Question 4. State important features of R.**Question 5.** Differentiate between vector, list, matrix, and data frame with general syntaxes and examples.**Question 6.** Write all R codes to perform correlation and regression analysis for the following data set:

Height (inches)	60	62	65	68	70	72
Weight (lbs)	110	120	140	160	170	190

Question 7. Write R syntaxes to apply t-test (single sample and two sample) and Chi-square test. For illustration, you may take data of your choice.**Question 8.** Explain the use of `next`, `break`, and `stop` functions. Give an example for each.**SECTION — C: Very Short Answer Questions**

[10 × 1 = 10]

Instructions: Answer the following questions in a single word or sentence.

- How do you enter integer, complex, character, and numeric data in R?
- Write R code to enter the following data in R using `seq()` and `rep()`:

1.0, 1.0, 1.0, 1.0, 1.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0

- Write general syntax of the import/export of a CSV file.
- Write a code to compute sum, mean, and product of a given vector of elements.
- Write R code to add a legend in R plot.
- How do you set the margin of the plot in R?
- R is an implementation of _____ language.
- Write R code to extract a column of a data frame.
- Write down the output of the codes:

```
N = 1:15
which(N >= 6 & N < 10)
```

- Write down the output of the codes:

```
x = letters[1:10]
x[seq(2, 10, by = 2)]
```

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